



Daily Practices Sheet - 11

MATHS

1. If the conditional $p \rightarrow q$ has truth value 'F' then which of the following proposition has truth value 'F'?
1) $\sim p \rightarrow q$ 2) $q \rightarrow p$ 3) $\sim p \rightarrow \sim q$ 4) $p \leftrightarrow q$
2. If $\sum_{i=1}^9 (x_i - 99) = 27$ and $\sum_{i=1}^9 (x_i - 99)^2 = 306$ then the mean of the squares of the deviations of $x_1, x_2, \dots, x_9$ from their A.M is
1) 9 2) 18 3) 25 4) 12
3. The sum of the digits of all the natural numbers from 1 to 300 is equal to
1) 3000 2) 3003 3) 3033 4) none of these
4. Let A be a set containing 32 elements. Let x_k denote the number of 'k' element sub-sets of A. Then $\sum_{i=0}^8 (x_{4i}) =$
1) $2^{30} + 2^8$ 2) $2^{29} + 2^8$ 3) $2^{15} (2^{15} + 1)$ 4) $2^{14} (2^{16} + 1)$
5. Mean deviation of the numbers 1, 2, 3, ..., $(2n + 1)$ from their median is
1) $\frac{n(n+1)}{2n+1}$ 2) $\frac{n(n-1)}{2n+1}$ 3) $\frac{n^2-1}{2n+1}$ 4) 0
6. If x, y are unit digit, 10's digit respectively of the number $(123)^{45}$, then the number of positive integral divisors of $(x^y \cdot y^x)$ is
1) 24 2) 35 3) 48 4) 60
7. Let $A = (a_{ij})_{3 \times 3}$ and $a_{ij} = \text{minimum of } \{i, j\}$. Then the trace of the matrix A^2 is equal to
1) 26 2) 28 3) 34 4) none of these
8. Let $a, b, c \in R$ and the system of equations $x + y + z = 0$, $ax + by + cz = 0$, $a^2x + b^2y + c^2z = 0$ has infinitely – many solutions. Then
1) $a = b = c$ 2) $a = b$ or $b = c$ or $c = a$
3) $a + b + c = 0$ 4) $a \neq b$ and $b \neq c$ and $c \neq a$

9. If the two expressions $99x^2 + 19x + 9$ and $55x^2 + ax + b$ have a common factor, then the value of fractional part of $(a + b)$ is ; $(a, b) \in \mathbb{R}$
 1) $2/9$ 2) $3/9$ 3) $4/9$ 4) $5/9$
10. If $l = \log_{0.5} x$ and $A = \{x \in N / x \text{ is prime and } l(l-1) < 30\}$ then $n(A) =$
 1) 9 2) 8 3) 10 4) 11
11. Let S_n, S_n^1 be the sum to 'n' terms of two series whose n^{th} terms are
 $t_n = (2n+1)\left(-\frac{1}{2}\right)^{n-1}$ and $t_n^1 = (2n-1)\left(\frac{1}{4}\right)^n$ respectively. Then
 $\lim_{n \rightarrow \infty} (S_n^1 - S_n) =$
 1) $\frac{20}{9}$ 2) $-\frac{20}{9}$ 3) 1 4) -1
12. If 9 H.Ms are inserted between $\frac{1}{36}$ and $\frac{1}{1296}$ then the A.M of the reciprocals of the third and the seventh H.Ms is equal to
 1) 630 2) 648 3) 666 4) 672
13. Let $D = \{x \in N / x \text{ is a divisor of } 2^3 \cdot 3^4 \cdot 5^5\}$. If a number is chosen at random from the elements of set D at random, then the probability that the number chosen is divisible by 5 or 6 is
 1) $\frac{14}{15}$ 2) $\frac{13}{15}$ 3) $\frac{4}{5}$ 4) $\frac{9}{10}$
14. Let $z_1 = 2 + 3i$ and $z_2 = 5 - i$ where $i = \sqrt{-1}$. If the curve traced out by the locus of 'z' such that $|z - z_1| + |z - z_2| = 20$ is inscribed in a rectangle, then the area of the rectangle is equal to $100\sqrt{k}$ square units, where $k =$
 1) 5 2) 10 3) 15 4) 2
15. The vectors $\vec{a}, \vec{b}, \vec{c}$ are such that the projection of $\vec{c}$ on $\vec{a}$ is equal to the projection of $\vec{c}$ on $\vec{b}$. If $|\vec{a}| = 2, |\vec{b}| = 1, |\vec{c}| = 3$ and $\vec{a} \cdot \vec{b} = 1$, then $|\vec{a} - 2\vec{b} - \vec{c}| =$
 1) $4\sqrt{3}$ 2) 6 3) $\sqrt{13}$ 4) $2\sqrt{6}$
16. The value of $\lim_{x \rightarrow 0} \frac{\sin(\sin(\pi \cos^2 x))}{x^2} =$
 1) 0 2) π 3) π^2 4) none of these

17. An arc of length 2π of the circle $x^2 + y^2 - 2x + 4y - c = 0$ is subtending an angle $\frac{\pi}{3}$ at the centre of the circle. If (a^2, b) lies on the circle and $b + 2 = 4\sqrt{2}$ then the integral part of $\left(\frac{c}{a^2}\right)$ is
1) 3 2) 6 3) 9 4) 10
18. If 3 distinct normals can be drawn from $(a, 0)$ to the parabola $y^2 = 12x$, then the least integral value of 'a' is
1) 7 2) 8 3) 9 4) 6
19. An ellipse with foci at $(\pm 2\sqrt{3}, 0)$ has its length of the latusrectum equal to 8 units. Then the product of the lengths of the perpendiculars drawn from the foci on any tangent to the ellipse is equal to
1) 16 2) 20 3) 24 4) 25
20. If the roots of the equation $x^2 - 4x + 1 = 0$ are the lengths of the semi-major axis and semi-minor axis of an ellipse, then the eccentricity of the ellipse lies between
1) $\frac{1}{3}$ and $\frac{1}{2}$ 2) $\frac{1}{4}$ and $\frac{1}{3}$ 3) $\frac{1}{2}$ and $\frac{2}{3}$ 4) $\frac{2}{3}$ and 1
21. Let A be the foot of the origin on the line $L : \frac{x-9}{1} = \frac{y}{2} = \frac{z}{-2}$ and B be the point $(12, 2, 2)$ then the length of the projection of $\overline{AB}$ on the line 'L' is equal to
1) 1 2) 2 3) 3 4) 4
22. A plane containing the line $\frac{x-1}{1} = \frac{y-3}{3} = \frac{z-5}{5}$ is parallel to the line $\frac{x-18}{1} = \frac{y-18}{2} = \frac{z-20}{3}$. Then which of the following points is not contained in that plane?
1) $(0, 0, 0)$ 2) $(4, 2, 0)$ 3) $(1, 2, 3)$ 4) $(-1, 2, 0)$
23. An observer finds the angle of elevation of a tower on a horizontal level plane as θ . On advancing 'a' metres towards the tower, the elevating is 45° ,

on further advancing 'b' meters, the elevation is $90^\circ - \theta$. The height of the tower is

- 1) $\frac{ab}{a+b}$ 2) $\frac{ab}{a-b}$ 3) $\frac{a-b}{ab}$ 4) $\frac{a+b}{ab}$

24. $\int \frac{dx}{\tan x + \cot x + \sec x + \operatorname{cosec} x} =$

- 1) $\frac{1}{2}(\sin x - x - \cos x) + c$ 2) $\frac{1}{2}(\cos x - x - \sin x) + c$
 3) $\frac{1}{2}(\sin x + x - \cos x) + c$ 4) $\frac{1}{2}(\cos x + x - \sin x) + c$

25. If $\{.\}$ denotes the fractional part function, then the value of $\int_0^9 \{\sqrt{x}\} dx =$

- 1) 4 2) 5 3) 6 4) 7

26. The area of the region bounded by the two curves

$y = \sqrt{1-x^2}$ and $y = (|x|-1)^4$ is $\left(\frac{\pi}{2} - \lambda\right)$ square units, then $\lambda =$

- 1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{4}{5}$ 4) $\frac{1}{10}$

27. The equation of family of curves whose differential equation is

$y^2 + \left(x - \frac{1}{y}\right) \frac{dy}{dx} = 0$ is given by

- 1) $x = \left(\frac{1}{y} + 1\right) - ce^{1/y}, c \in R$ 2) $x = \left(\frac{1}{y} - 1\right) + ce^{1/y}, c \in R$
 3) $x = \left(\frac{1}{y} + 1\right) + ce^{-1/y}, c \in R$ 4) $x = \left(\frac{1}{y} - 1\right) + ce^{-1/y}, c \in R$

28. The set of all triplets of real numbers (a, b, c) with $a \neq 0$, such that the

function $f(x) = \begin{cases} x & \text{if } x \leq 1 \\ ax^2 + bx + c & \text{if } x > 1 \end{cases}$ is differentiable for all real 'x' is

given by

- 1) $\{(t, 1-2t, t) / t \in R, t \neq 0\}$ 2) $\{(t, 2t-1, t) / t \in R, t \neq 0\}$
 3) $\{(t, 1-2t, 2t) / t \in R, t \neq 0\}$ 4) $\{(t, 2t-1, 2t) / t \in R, t \neq 0\}$

29. The derivative of $f(\tan x)$ with respect to $g(\sec x)$ at

$$x = \frac{\pi}{4}, \text{ if } f'(1) = 2 \text{ and } g'(\sqrt{2}) = 4 \text{ is}$$

- 1) 2 2) $2\sqrt{2}$ 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{1}{2\sqrt{2}}$

30. If $(2, -1)$ is a turning point of the curve $y = f(x)$ where $f(x) = \frac{ax+b}{(x-1)(x-4)}$

then local minimum value of $f(x)$ is

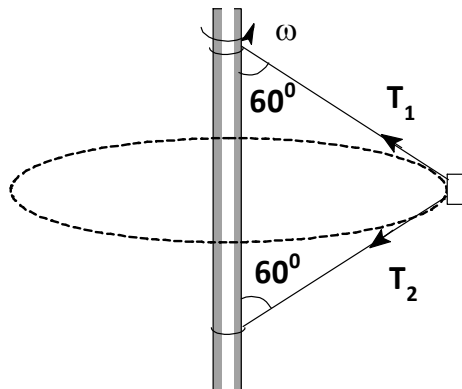
- 1) $-\frac{1}{3}$ 2) $\frac{1}{4}$ 3) $-\frac{1}{6}$ 4) $-\frac{1}{9}$

PHYSICS

31. At a certain height a body at rest explodes into two equal fragments with two fragments receiving horizontal velocities each equal to 10ms^{-1} in opposite directions. The time interval after the explosion for which the velocity vectors of the two fragments become perpendicular to each other ($g = 10\text{ms}^{-2}$)

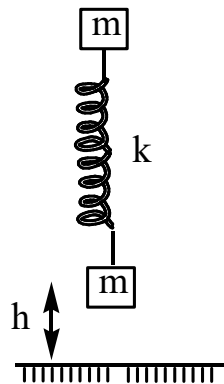
- 1) 1s 2) 2s 3) 1.5s 4) 1.75s

32. A small block is connected to one end of two identical massless strings of length $16\frac{2}{3}\text{cm}$ each with their other ends fixed to a vertical rod. If the ratio of tensions $\frac{T_1}{T_2}$ be 4 : 1 then the angular velocity ω of the block will be:

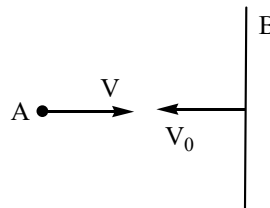


- 1) 10rads^{-1} 2) 9.8 rads^{-1} 3) 14 rads^{-1} 4) 19.6 rads^{-1}

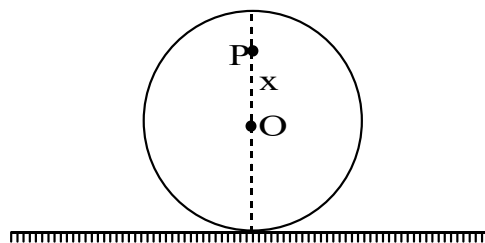
33. A spring mass system is held at rest with the spring relaxed at a height h above the ground. The minimum value of h for which the system has a tendency to rebound after hitting the ground is (assume zero coefficient of restitution for lower block and ground)



- 1) $\frac{2mg}{k}$ 2) $\frac{3mg}{k}$ 3) $\frac{3mg}{2k}$ 4) $\frac{6mg}{k}$
34. A body A moves towards a wall with velocity V . The wall B also moves towards the body A with velocity V_0 . After collision, the body moves in opposite direction with velocity V' which is $\left(1 + \frac{2V_0}{V}\right)$ times the velocity V . Find the coefficient of restitution:

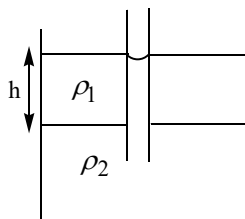


- 1) $\frac{1}{4}$ 2) $\frac{1}{3}$ 3) 1.0 4) $\frac{1}{2}$
35. A disc of radius R is rolling without slipping on a horizontal surface. P is a point at distance x above its centre O at a certain instant. The radius of curvature of the trajectory of P at that instant is



- 1) $\frac{(R+x)^2}{x}$ 2) $R+x$ 3) $\frac{(R+x)^2}{R}$ 4) $\frac{R^2}{R-x}$

36. A container has two immiscible liquids of densities ρ_1 and $\rho_2 (> \rho_1)$. A capillary tube of radius r is inserted in the liquid so that its bottom reaches up to the denser liquid. The denser liquid rises in the capillary and attains a height 'h' from the interface of the liquids which is just equal to length of the lighter liquid. Assuming angle of contact to be zero the surface tension of heavier liquid is



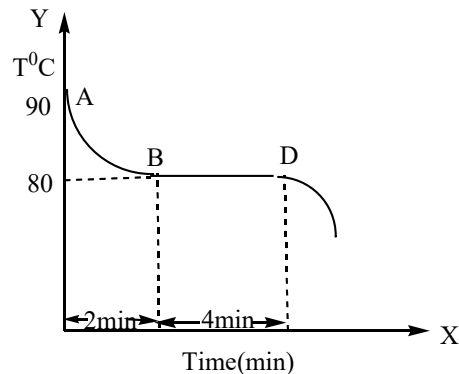
- 1) $2\pi r \rho_2 gh$ 2) $\frac{\rho_2 gh}{2}$ 3) $\frac{r}{2}(\rho_2 - \rho_1)gh$ 4) $2\pi r(\rho_2 - \rho_1)gh$
37. A projectile is projected with a velocity $\frac{3}{4}V_e$ vertically up from the surface of the earth where V_e is the escape velocity on the surface of the earth. The maximum height attained by the projectile from the surface of the earth will be (R_e is the radius of earth)
- 1) $\frac{9}{16}R_e$ 2) $\frac{9R_e}{7}$ 3) R_e 4) $\frac{6R_e}{7}$
38. A satellite in an equatorial orbit has a time period of 6 hrs. At a certain instant, it is directly overhead of an observer on the equator of the earth. It is directly overhead the observer again after a time T. The possible value of T is/are
- 1) 8 hr 2) 4.8 hr
3) Both (1) and (2) 4) none of these
39. In an electron gun, emission current from the cathode is given by the equation, $I = AT^2 e^{-B/(KT)}$ [K = Boltzmann constant, A = constant]. The dimensional formula for AB^2 is same as
- 1) KT 2) IT^2 3) IK^2 4) $\frac{IK^2}{T}$

40. At what temperature is the root mean square speed of an atom in an argon gas cylinder equal to the rms speed of a helium gas atom at -20°C ?
(Atomic mass of $Ar = 39.9\text{u}$ & of $He = 4.0\text{u}$)

1) $3.52 \times 10^3\text{K}$ 2) $2.52 \times 10^3\text{K}$ 3) $2.52 \times 10^4\text{K}$ 4) $3.52 \times 10^4\text{K}$

41. The figure given below shows the cooling curve of pure wax material after heating. It cools from A to B and solidifies along BD. If L and C are respectively values of latent heat and the specific heat of the liquid wax, the ratio L/C is

(Surrounding temperature is 60°C and consider only radiation)



1) 4 2) 16 3) 24 4) 48

42. A vessel is partly filled with liquid. When the vessel is cooled to a lower temperature, the space in the vessel, unoccupied by the liquid remains constant. Then the volume of the liquid (V_L), volume of the vessel (V_v), the coefficients of cubical expansion of the material of the vessel (γ_v) and that of the liquid (γ_L) are related as:

1) $\gamma_L > \gamma_v$ 2) $\gamma_v / \gamma_L = V_L / V_v$
3) $\gamma_v / \gamma_L = V_v / V_L$ 4) Both 1 and 2

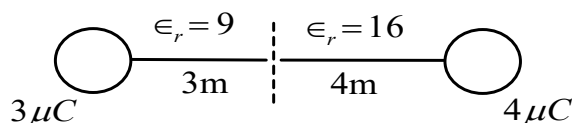
43. A wire of radius 0.5 mm is held tight under tension 100 N on two edges separated by a distance of 50 cm. In its fundamental mode of vibration it produces 6 beats per second with a tuning fork. When the tension is reduced to 81 N, the wire produces beats at the same rate with the same tuning fork. The frequency of the tuning fork is

1) 85 Hz 2) 114 Hz 3) 95 Hz 4) 104 Hz

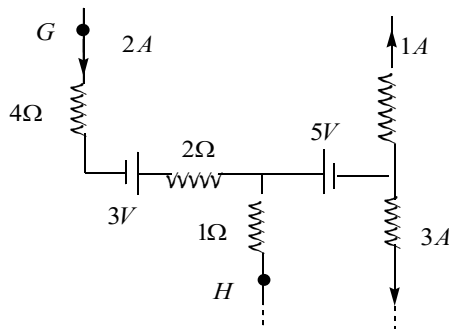
44. A boat is travelling in a river with a speed 10ms^{-1} along the stream flowing with a speed 2ms^{-1} . From this boat, a sound transmitter is lowered into the river through a rigid support. The wavelength of the sound emitted from the transmitter inside the water is 14.45mm . Assume that attenuation of sound in water and air is negligible.

What will be the frequency of sound in water (Bulk modulus of water = $2.088 \times 10^9 \text{ pa}$)

- 1) 10^5 Hz 2) 10^6 Hz 3) 10^7 Hz 4) 10^8 Hz
45. The point charges $3\mu\text{C}$ and $4\mu\text{C}$ are placed at a separation of 7m . The medium between them is of two types as shown in figure, the electric force acting between them is



- 1) $1.75 \times 10^{-3} \text{ N}$ 2) $1.73 \times 10^{-4} \text{ N}$ 3) $0.18 \times 10^{-4} \text{ N}$ 4) $0.18 \times 10^{-3} \text{ N}$
46. In the part of a circuit show in figure, the potential difference ($V_H - V_G$) between points G and H will be

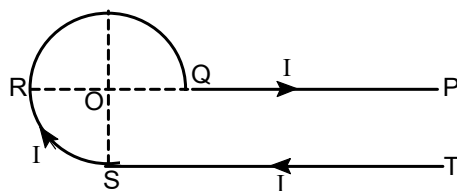


- 1) 1 V 2) -1 V 3) 7 V 4) -7 V
47. A capacitor is charged to a certain potential and then allowed to discharge through a resistance R , the ratio of charge on the capacitor to current in the circuit
- 1) changes with time
 - 2) does not change with time and it is equal to time constant of circuit
 - 3) does not change with time, and is equal to half of the time constant of circuit
 - 4) may or may not change depending upon the charge given to the capacitor

48. A charged particle is released from rest in a region of uniform electric and magnetic fields, which are parallel to each other. The locus of the particle will be

- 1) helix of constant pitch 2) straight line
3) helix of varying pitch 4) cycloid

49. Current I is flowing in a conductor shaped as shown in fig. The radius of the curved part is r and the length of the straight portion is very large. The value of the magnetic field at the centre O will be



- 1) $\frac{\mu_0 I}{4\pi r} \left(\frac{\pi}{2} + 1 \right)$ 2) $\frac{\mu_0 I}{4\pi r} \left(\frac{\pi}{2} - 1 \right)$ 3) $\frac{\mu_0 I}{4\pi r} \left(\frac{3\pi}{2} + 1 \right)$ 4) $\frac{\mu_0 I}{4\pi r} \left(\frac{3\pi}{2} - 1 \right)$

50. An alternating current $I = I_0 \sin \omega t$ is flowing in a circuit. The ratio of rms current in the interval of 0 to T , the average current in the circuit for the time interval from $T/8$ to $3T/8$ (Where “ T ” is time period) is :

- 1) $\frac{\pi}{2\sqrt{2}}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\frac{\sqrt{2}\pi}{4}$

51. A combination of two thin lenses with focal lengths f_1 and f_2 respectively forms a real image of distant object behind the rear lens (f_2) at a distance 60cm from f_2 when lenses are in contact. The position of this image shifts by 30cm towards the combination when two lenses are separated by 10cm by moving f_1 towards the object. The corresponding values of f_1 and f_2 are

- 1) 30cm, -60cm 2) 20cm, -30cm
3) 15cm, -20cm 4) 12cm, -15cm

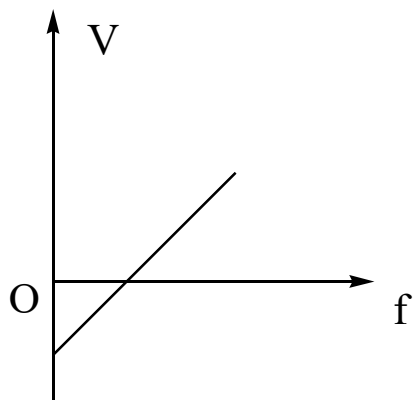
52. When photon of energy 4.25 eV strike the surface of a metal A, the ejected photo electrons have maximum kinetic energy $T_A eV$ and de – Broglie wave length λ_A . The maximum kinetic energy of photo electrons liberated from another metal B by photon of energy 4.7 eV is $T_B = (T_A - 1.5) eV$. If the de-Broglie wave length of these photo electrons is $\lambda_B = 2\lambda_A$ Then

- 1) Work function of A is 2.25 eV 2) Work function of B is 7.26 eV
 3) $T_A = 4 eV$ 4) $T_B = 6 eV$

53. The Maxwell's equation $\oint \vec{B} \cdot d\vec{l} = \mu_0 \left[I + \epsilon_0 \frac{d\phi_E}{dt} \right]$ is a statement of

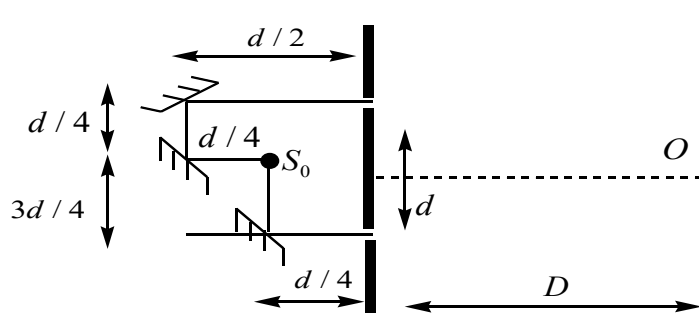
- 1) Faraday's law of induction 2) Modified Ampere's Law
 3) Gauss's law of electricity 4) Gauss's law of magnetism

54. The stopping potential V for photo electric emission from a metal surface is plotted along y – axis and frequency 'f' of incident light along x-axis. A straight line is obtained as shown. Plank's constant is given by

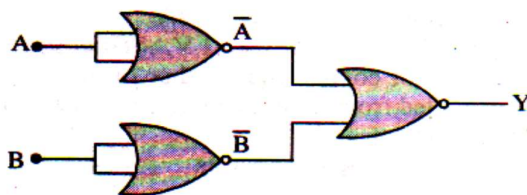


- 1) Slope of the line
 2) Product of the slope of the line and charge of the electron
 3) Product of intercept along y – axis and mass of the electron
 4) Product of slope and mass of electron

55. In a YDSE apparatus, the source (wavelength λ) is kept as shown. None of the slits receives direct light but get it after reflection by mirrors as shown in the figure. If $d = 100\lambda$ and $D = 50000\lambda$. Then the distance of central maxima from the centre of the screen

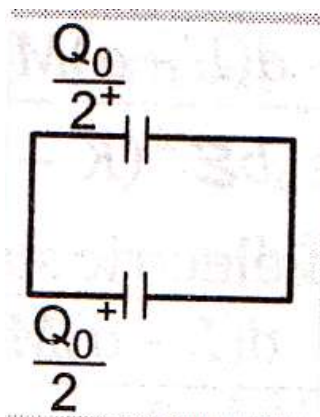


- 1) 100λ 2) 200λ 3) 250λ 4) 500λ
56. Neon-23 decays in the following way ${}_{10}\text{Ne}^{23} \rightarrow {}_{11}\text{Na}^{23} + {}_{-1}\text{e}^0 + \bar{\nu}$. The maximum KE that the beta particle (${}_{-1}\text{e}^0$) can have [The atomic masses of $\text{Ne}^{23} = 22.9945\text{u}$ and for $\text{Na}^{23} = 22.9895\text{u}$] (1 amu = 931.5 MeV)
- 1) 4.66 MeV 2) Zero 3) 0.2 MeV 4) 8.2 MeV
57. Identify the operation performed by the circuit given below



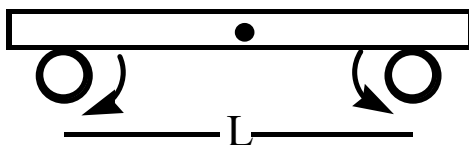
- 1) NOT 2) AND 3) OR 4) NAND
58. Which of the following are different blocks of detection circuit of Amplitude modulated wave
- A: square law device B: BAND pass filter
C: Rectifier D: Envelope detector
- 1) A only 2) B only 3) Both A & B 4) Both C & D
59. Two capacitors each having area A and plate separation 'd' are connected as shown in the circuit. Each capacitor carries charge $\frac{Q_0}{2}$. The plates of one capacitor are slowly pulled apart by an external agent till the separation

between them becomes $2d$. The other capacitor is not disturbed. The work done by the external agent in this process will be :



- 1) $\frac{Q_0^2 d}{12A\epsilon_0}$ 2) $\frac{Q_0^2 d}{6A\epsilon_0}$ 3) $\frac{Q_0^2 d}{18A\epsilon_0}$ 4) $\frac{Q_0^2 d}{9A\epsilon_0}$

60. A uniform plate of mass m stays horizontally and symmetrically on two wheels rotating in opposite directions. The separation between the wheels is L . The friction coefficient between each wheel and the plate is μ . Find the time period of oscillations of the plate if it is slightly displaced along its length and released.



- 1) $2\pi\sqrt{\frac{L}{2\mu g}}$ 2) $2\pi\sqrt{\frac{L}{\mu g}}$ 3) Zero 4) $2\pi\sqrt{\frac{L}{g}}$

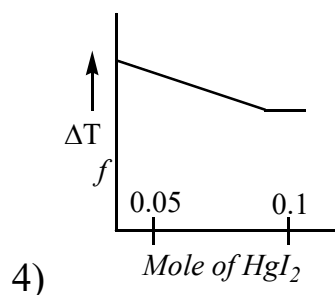
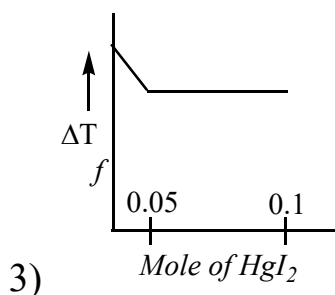
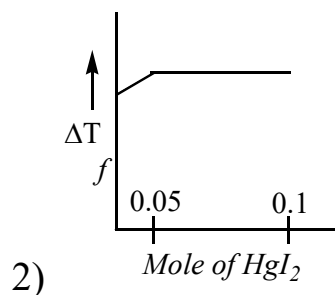
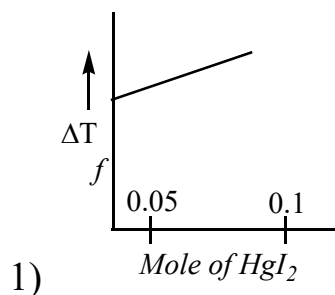
CHEMISTRY

61. The number of radial nodes present in $3p$ orbital is
 1) 0 2) 1 3) 2 4) 3
62. A gas container observes Maxwell distribution of speeds. If the number of molecules between the speed 5 and 5.1 km sec^{-1} at 25°C be 'N', what would be the number of molecules between this range of speed if the total number of molecules in the vessel are doubled?
 1) N^2 2) N 3) $N/2$ 4) $2N$

63. Lead (II) azide enters into a redox reaction with cobalt (III) manganate as follows: $Pb(N_3)_2 + Co(MnO_4)_3 \rightarrow MnO_2 + CoO + NO + Pb_3O_4$

What is the number of moles of electrons transferred when 3 moles of $Pb(N_3)_2$ is oxidized.

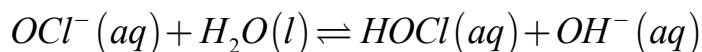
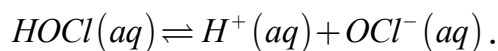
- 1) 9 moles 2) 14 moles 3) 23 moles 4) 44 moles
64. Increasing amount of HgI_2 is added to 1 lit of an aqueous solution containing 0.1 mole of KI. Which of the following graphs does represent the variation of depression in freezing point of the resulting solution with the amount of HgI_2 added?



65. A certain radioactive element A, has a half-life = t seconds. In $(t/2)$ seconds the fraction of the initial quantity of the element so far decayed is nearly

- 1) 25% 2) 29% 3) 21% 4) 17%
66. Given the following reaction at equilibrium $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$. Some inert gas is added at constant Volume predict which of the following facts will be affected?
- 1) More of $NH_{3(g)}$ is produced 2) less of $NH_{3(g)}$ is produced
- 3) No affect on the degree of advancement of the reaction at equilibrium
- 4) K_p of the reaction is increased

67. Hypochlorous acid ionizes as



K_a for this reaction at $25^\circ C$ is 3.0×10^{-8} ($K_w = 1.0 \times 10^{-14}$ at $25^\circ C$)

Hence, K_b for OCl^- is

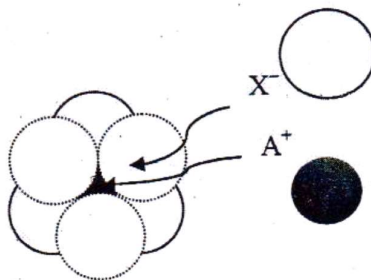
- 1) 3.3×10^{-7} 2) 3.0×10^{-8} 3) 3.0×10^6 4) 3.3×10^7

68. One mole of an ideal gas $\left(C_{V,m} = \frac{5}{2}R \right)$ at 300 K 5 atm is expanded

adiabatically to a final pressure of 2 atm against a constant pressure of 2 atm. Final temperature is

- 1) 270K 2) 273K 3) 248.5K 4) 200K

69. The arrangement of X^- ions around A^+ ion in solid AX is given in the figure (not drawn to scale). If the radius X^- of is 250 pm, the radius of A^+ is



- 1) 104 pm 2) 125pm 3) 183 pm 4) 57pm

70. The reduction potential of the hydrogen half-cell will be negative if

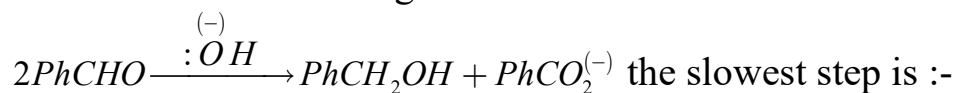
- 1) $p(H_2) = 1$ atm and $[H^+] = 1M$ 2) $p(H_2) = 1$ atm and $[H^+] = 2M$
 3) $p(H_2) = 2$ atm and $[H^+] = 1M$ 4) $p(H_2) = 2$ atm and $[H^+] = 2M$

71. $Cl_2 + 2NaOH(\text{cold / dil}) \rightarrow \dots + \dots + \dots$

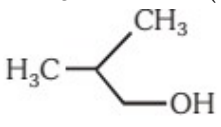
What is the O.N. of Cl in anion of oxyacid in product

- 1) +3 2) +1 3) +5 4) -1

72. In Cannizzaro reaction given below :-



- 1) The abstraction of proton from the carboxylic group
 2) The deprotonation of $PhCH_2OH$

- 3) The attack of : $\overset{(-)}{O}H$ at the carboxyl group
 4) The transfer of hydride to the carbonyl group
73. Among the following the one that gives positive iodoform test upon reaction with I_2 and NaOH is-
- 1) $CH_3CH_2CH(OH)CH_2CH_3$ 2) $C_6H_5CH_2CH_2OH$
- 3)  4) $PhCHOHCH_3$
74. The hybridisation of orbitals of N-atom in NO_3^- , NO_2^+ and NH_4^+ are respectively:-
- 1) sp, sp^3, sp^2 2) sp^2, sp^3, sp 3) sp, sp^2, sp^3 4) sp^2, sp, sp^3
75. One mole of the complex compound $Co(NH_3)_5Cl_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of $AgNO_3$ solution to yield two moles of $AgCl$ (s). The structure of the complex is
- 1) $[Co(NH_3)_3Cl_3] \cdot 2NH_3$ 2) $[Co(NH_3)_4Cl_2]Cl \cdot NH_3$
 3) $[Co(NH_3)_5]Cl_3$ 4) $[Co(NH_3)_5Cl]Cl_2$
76. The correct order of magnetic moments (spin only values in B.M.) among is :-
- 1) $[Fe(CN)_6]^{4-} > [MnCl_4]^{2-} > [CoCl_4]^{2-}$
 2) $[MnCl_4]^{2-} > [Fe(CN)_6]^{4-} > [CoCl_4]^{2-}$
 3) $[MnCl_4]^{2-} > [CoCl_4]^{2-} > [Fe(CN)_6]^{4-}$
 4) $[Fe(CN)_6]^{4-} > [CoCl_4]^{2-} > [MnCl_4]^{2-}$
 (Atomic nos. : Mn = 25, Fe = 26, Co = 27)
77. Aluminium is extracted by the electrolysis of :
- 1) Bauxite
 2) Alumina
 3) Alumina mixed with molten cryolite
 4) Molten cryolite
78. The structure of diborane contains
- 1) Four 2c-2e bonds and four 3c-2e bonds
 2) Two 2c-2e bonds and two 3c-2e bonds
 3) Two 2c-2e bonds and four 3c-2e bonds

4) Four 2c-2e bonds and two 3c-2e bonds

79. Identify the incorrect statement among the following

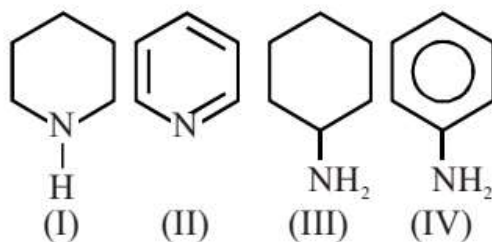
1) Ozone reacts with SO_2 to give SO_3

2) Silicon reacts with $NaOH(aq.)$ to give Na_2SiO_3 and H_2

3) Cl_2 reacts with excess of NH_3 to give N_2 and HCl

4) Br_2 reacts with hot and strong $NaOH$ solution to give $NaBr$, $NaBrO_4$ and H_2O

80. Rank the following in increasing basic nature :



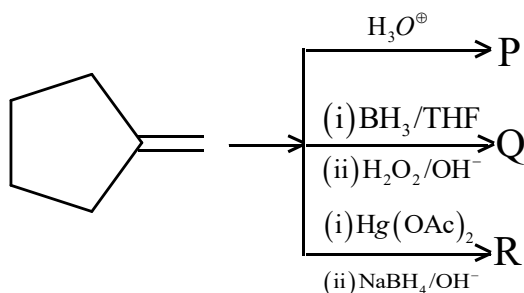
1) $III > I > II > IV$

2) $I > III > II > IV$

3) $IV > II > III > I$

4) $I > III > IV > II$

81.



Which is correct for product, P, Q and R (P, Q, R are major product)

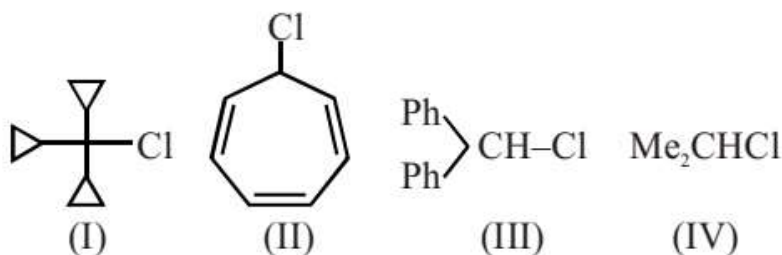
1) Product P & R are identical

2) Product Q & R are identical

3) Product P & Q are F.G. isomers

4) Product P, Q & R are different

82. Correct order of rate of SN^1 :

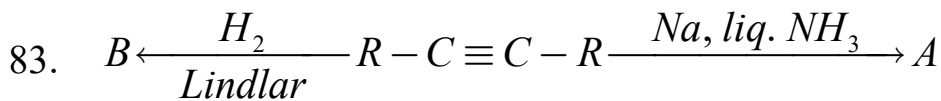


1) $I > II > III > IV$

2) $III > II > I > IV$

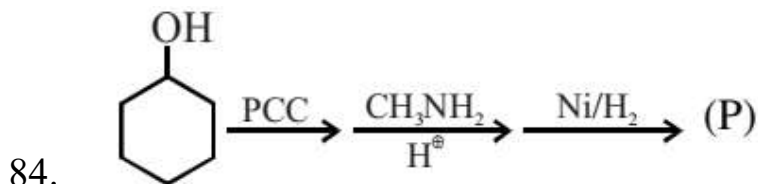
3) $I > III > II > IV$

4) $III > I > II > IV$

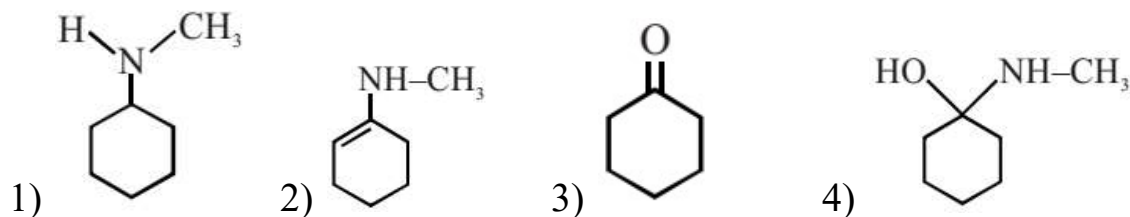


A and B are geometrical isomers :-

- 1) A is trans, B is cis 2) A and B both are cis
3) A and B both are trans 4) A is cis, B is trans



P will be :



85. According to MO theory which of the list ranks the oxygen species in terms of increasing Bond order $O_2, O_2^+, O_2^-, O_2^{2-}$

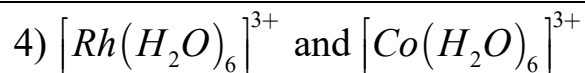
- 1) $O_2^{2-}, O_2^-, O_2^+, O_2$ 2) $O_2^+, O_2, O_2^-, O_2^{2-}$
3) $O_2, O_2^+, O_2^-, O_2^{2-}$ 4) $O_2^{2-}, O_2^-, O_2, O_2^+$

86. CrO_4^{2-} and MnO_4^- are strong yellow and intense purple respectively in aqueous solution. The colour is due to

- 1) charge transfer 2) $d-d$ transition
3) half-filled d -sub-shells 4) increasing number of unpaired electron

87. Among the following pairs of complexes, in which case Δ_0 value is higher for the first one?

- 1) $[Co(NH_3)_6]^{3+}$ and $[Co(CN)_6]^{3-}$
2) $[CoF_6]^{3-}$ and $[Co(NH_3)_6]^{3+}$
3) $[Co(H_2O)_6]^{2+}$ and $[Co(H_2O)_6]^{3+}$



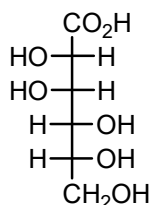
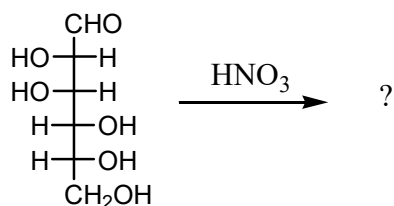
88. Which of the following is incorrect?

- 1) PbS----- Black ppt 2) CuS-----Black ppt
 3) CdS----Yellow ppt 4) MnS----- Green ppt

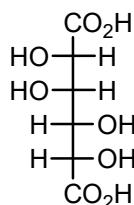
89. The product of addition polymerization reaction is

- 1) PVC 2) nylon 6 6
 3) terylene 4) nylon 2 nylon 6

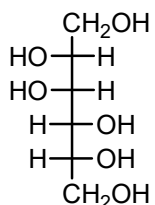
90. Reaction of the following substance with nitric acid would yield:



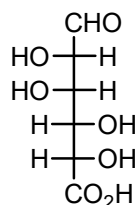
I



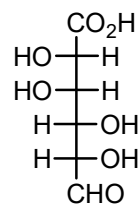
II



III



IV



V

1) III

2) IV & V

3) II

4) I

KEY SHEET

1	4	2	3	3	2	4	3	5	1
6	2	7	1	8	2	9	4	10	4
11	4	12	3	13	1	14	3	15	3
16	2	17	4	18	1	19	3	20	4
21	4	22	4	23	2	24	1	25	2
26	2	27	1	28	1	29	3	30	4
31	1	32	3	33	3	34	3	35	1
36	3	37	2	38	3	39	3	40	2
41	2	42	4	43	2	44	1	45	2
46	4	47	2	48	2	49	3	50	2
51	2	52	1	53	2	54	2	55	3
56	1	57	2	58	4	59	1	60	1
61	2	62	4	63	4	64	3	65	2
66	3	67	1	68	3	69	1	70	3
71	2	72	4	73	4	74	4	75	4
76	3	77	3	78	4	79	4	80	2
81	1	82	1	83	1	84	1	85	4
86	1	87	4	88	4	89	1	90	3